

The total possible points are 75, plus 10 points bonus for success in solving all values in Problem 2.69.

P1 (15): (a) 5 points; (b) 5 points; (c) 5 points.

P2 (25): (a) 14 points, each value of the variables worth 2 points; (b) 5 points; (c) 6 points.

P3 (15): DOF worth 6 points; realizing of overspecified conditions worth 4 points; proposing a reasonable adaptation of the system worth 5 points. A "reasonable adaptation" can be adding more operation units, relaxing current constraints, using soft limits, etc.

P4 (20 + 10): DOF worth 8 points; all equations for all operation units worth 12 points. For bonus part (which is to solve the equations), it is a non-linear equation system therefore could have multiple solutions and cannot be solved in a matrix. I highly recommend you to use a commercially available package to solve it such as Excel, Matlab or Python; another way is to use online tools like Wolfram. In the end, you should be able to eliminate physically impossible solutions which include negative values and be left with the only possible solution.